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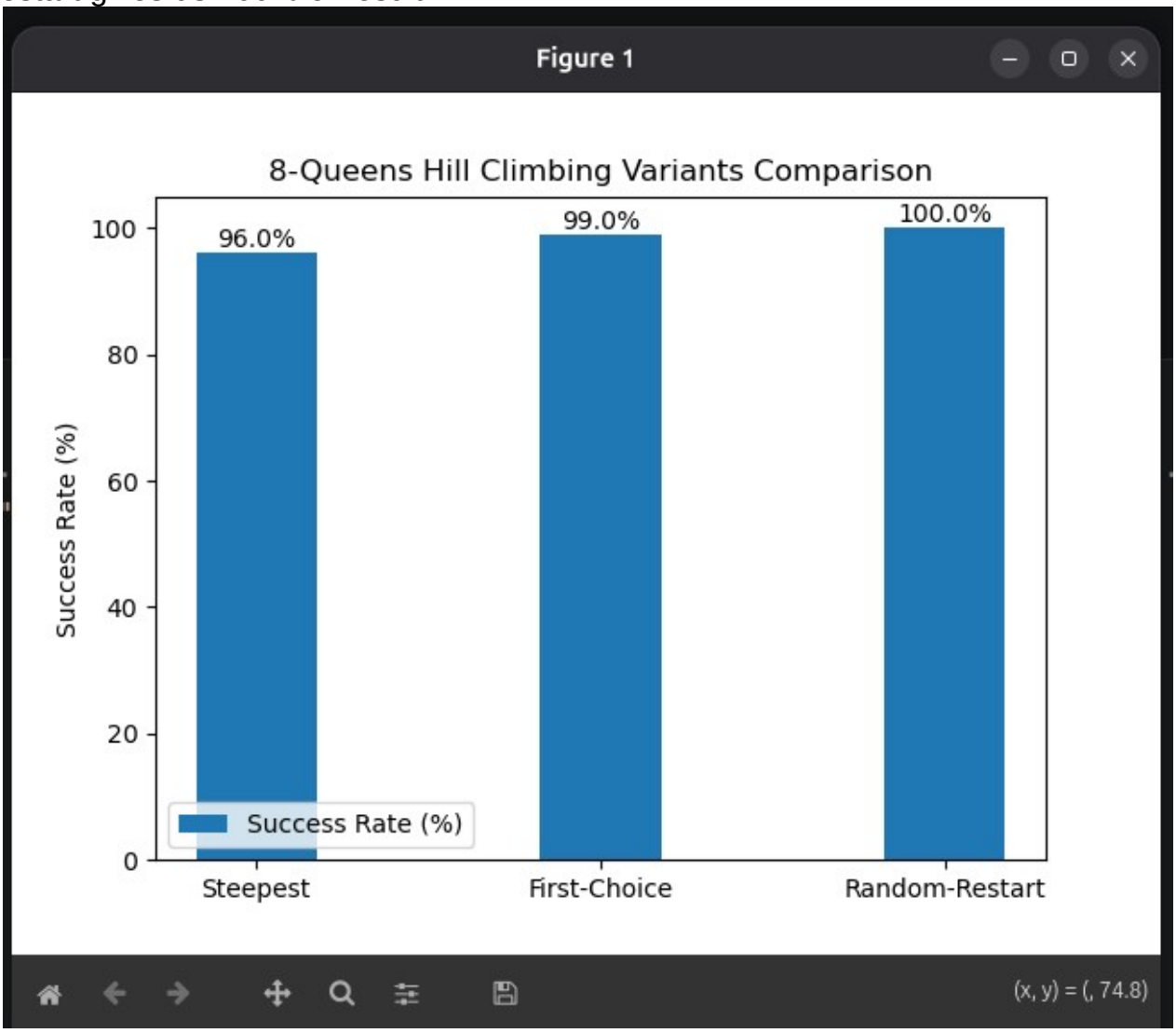
section : BS-CS-6A
Lab task : local searches

TASK 01 — N-QUEENS WITH HILL CLIMBING

output screen shot
Bar chart comparing steepest , first choice and random restart in which random restart gives us 100% of result



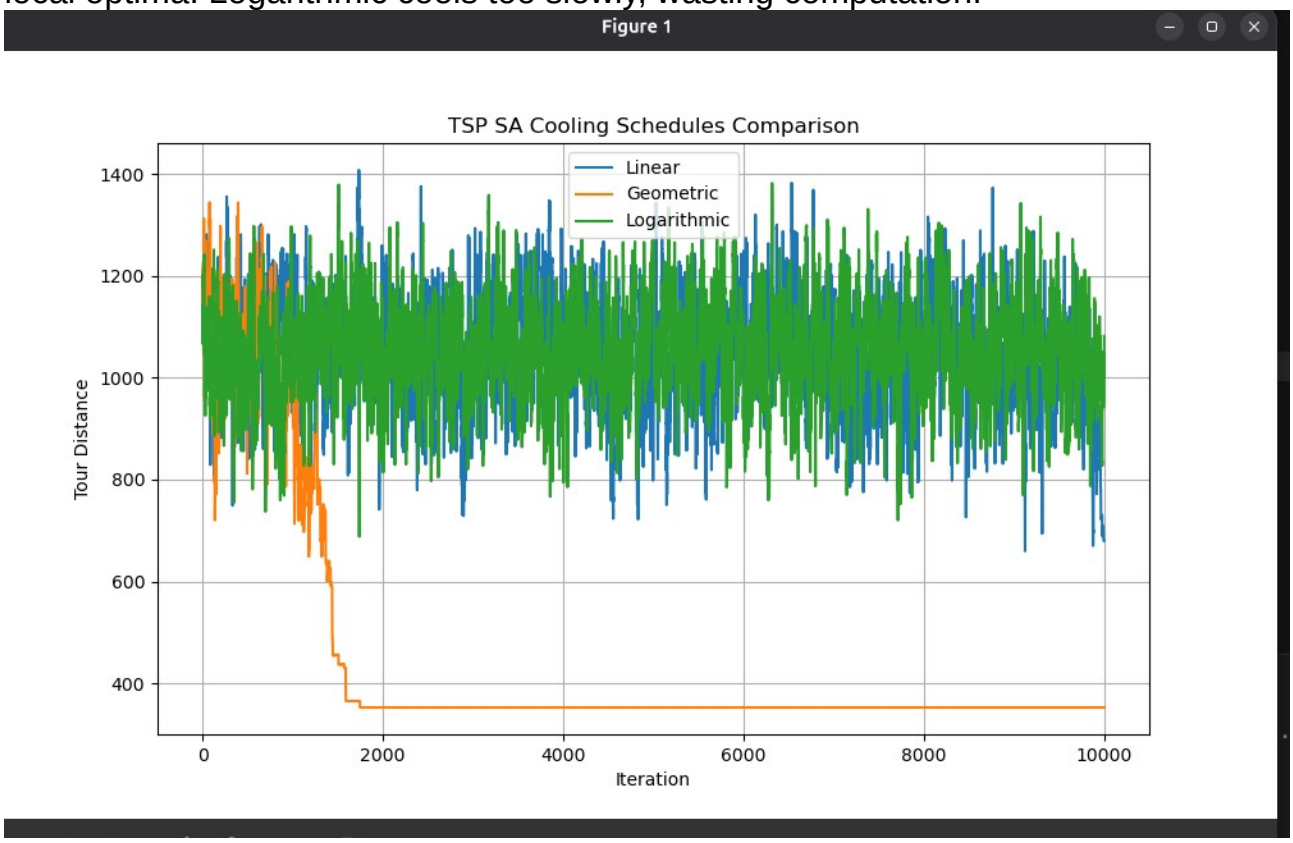

```
~/Desktop/AI_LAb/week_7_local_searchs  on  main ?2
python -u "/home/kalamay/Desktop/AI_LAb/week_7_local_searchs/q1.py"
Steepest success rate: 100.00%
First-choice success: 97.00%, avg time: 0.38ms
Random-restart success: 100.00%, avg restarts: 1.0
```

TASK 02 — TSP WITH SIMULATED ANNEALING

output screen shot

result :

Geometric cooling with T0=10000 and alpha=0.995 performs best. It allows initial exploration with high T and gradual exploitation. Linear cools too quickly, risking local optima. Logarithmic cools too slowly, wasting computation.



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```
~/Desktop/AI_LAb/week_7_local_searchs  on  main ?2
python -u "/home/kalamay/Desktop/AI_LAb/week_7_local_searchs/q2.py"
Running SA...
Initial distance: 1180.88
Linear SA: 679.63
Geometric SA: 352.32
Log SA: 1082.13
```

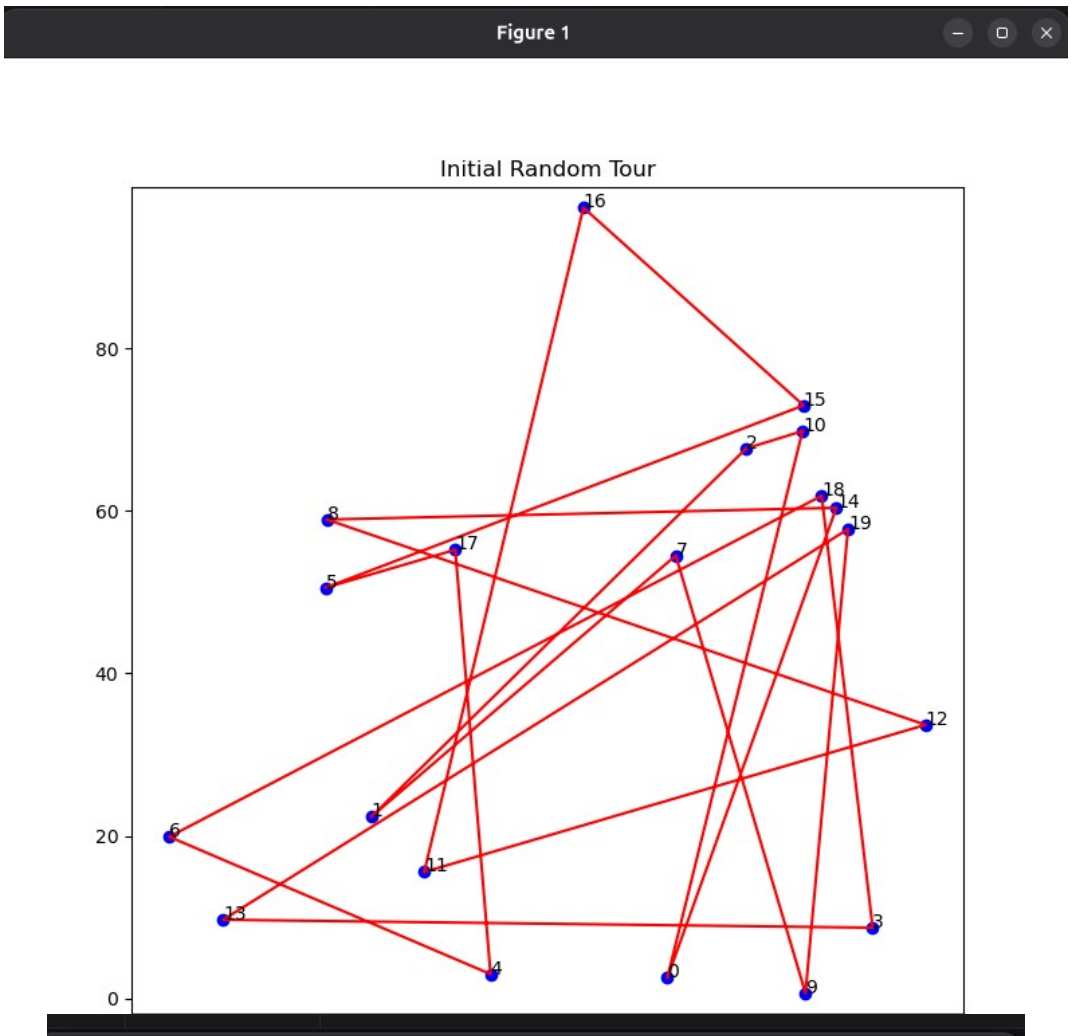


Figure 1
Initial Random Tour
16
15
10
2
18
14
19
8
17
7
5
12
1
6
11
13
3
4
0
9
0
20
40
60
80

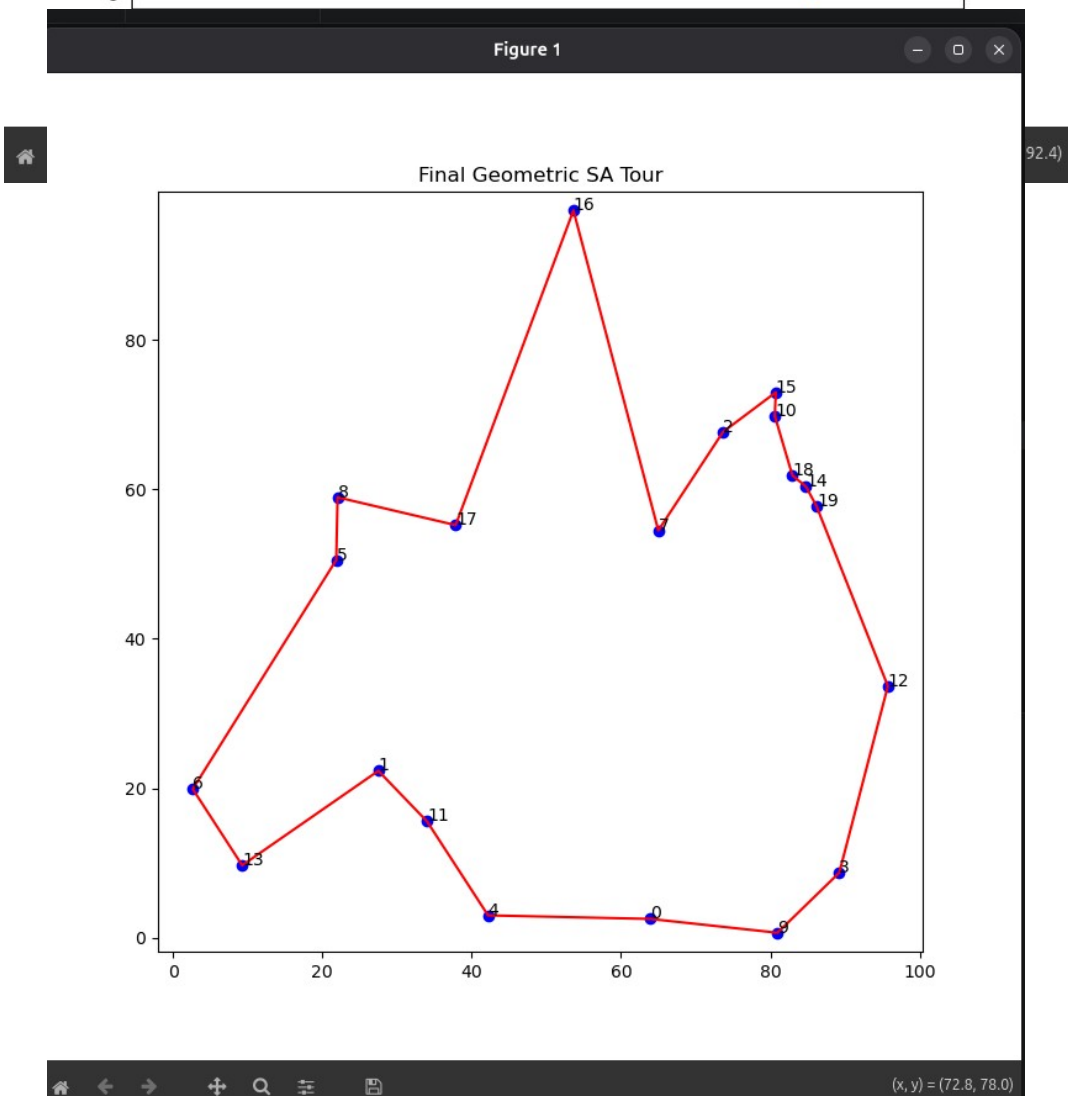
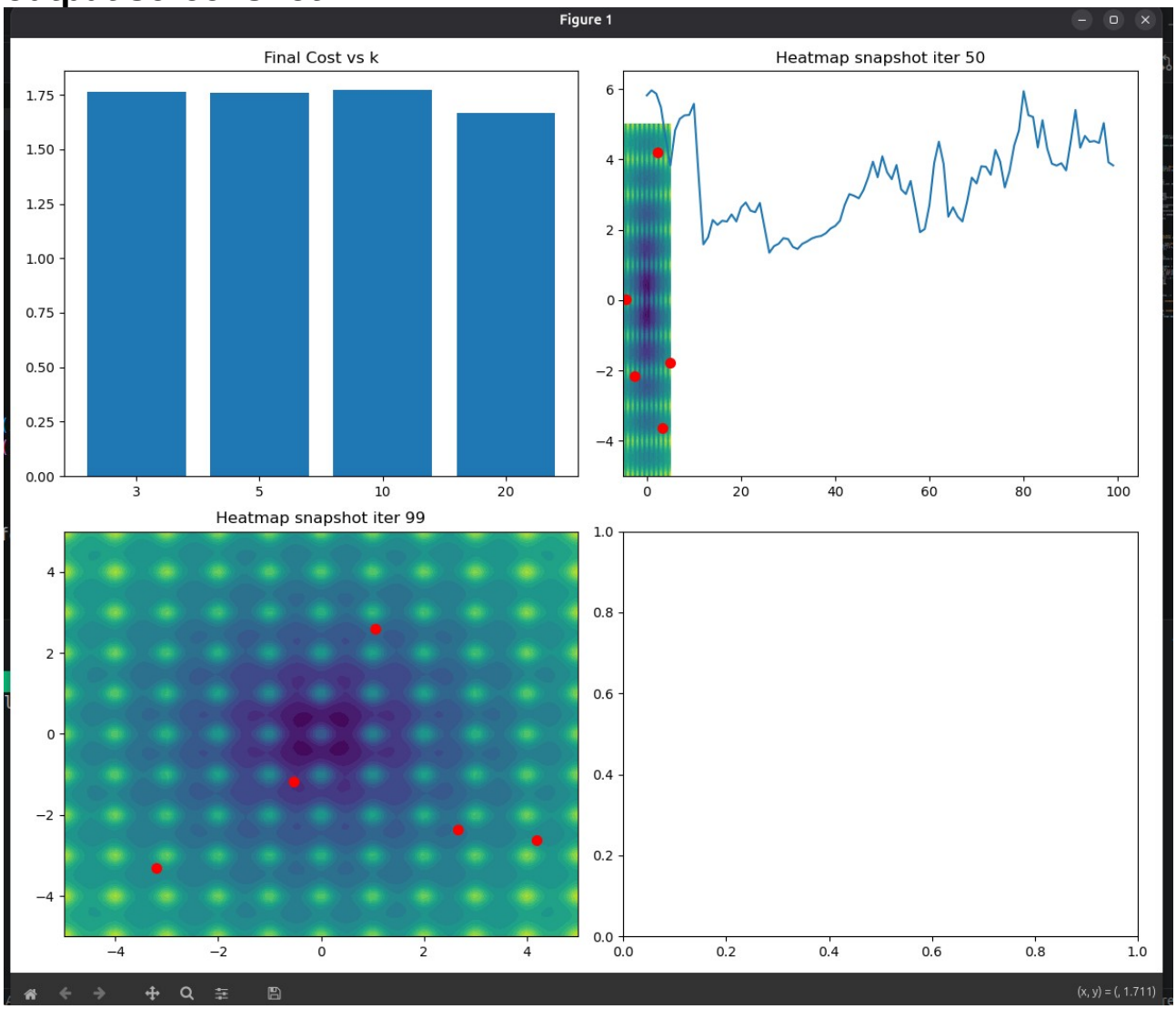


Figure 1
Final Geometric SA Tour
16
15
10
2
18
14
19
8
17
7
5
12
1
6
11
13
3
4
0
9
0
20
40
60
80
100
(x, y) = (72.8, 78.0)

TASK 03 — ACKLEY FUNCTION WITH BEAM SEARCH

output screenshot




```
~/Desktop/AI_LAb/week_7_local_searchs on main ?2
python -u "/home/kalamay/Desktop/AI_LAb/week_7_local_searchs/q3.py"
Running beam search for Ackley...
k=3: 1.7642
k=5: 1.7571
k=10: 1.7704
k=20: 1.6658
k=10+ diminishing returns: larger beams converge to same local mins (diversity loss).

~/Desktop/AI_LAb/week_7_local_searchs on main ?2
```

TASK 04 — JOB SCHEDULING WITH TABU SEARCH

output screen shot
result:

Random: 20.0
Best schedule Gantt:
Aspiration allows escaping tabu traps when improving global best. Triggered in runs with longer tenure.

```
~/Desktop/AI_LAb/week_7_local_searchs on main ?2
python -u "/home/kalamay/Desktop/AI_LAb/week_7_local_searchs/q4.py"
JSSP Tabu Search Results:
Tenure | Makespan | Aspiration
     3 |     27.0 | False
     7 |     27.0 | False
    15 |     24.0 | True
Random: 20.0
Best schedule Gantt:
Aspiration allows escaping tabu traps when improving global best. Triggered in runs with longer tenure.
```